SLIDE-QUIZ EXERCISE · EVOLLAND01

Evolution of Terrestrial Life

The Colonization of Land Over 500 Million Years

STUDENT STUDY GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EXERCISE OVERVIEW

Evolution of Terrestrial Life

This exercise follows life's conquest of Earth's land surface across 500 million years, from microscopic crusts on bare rock to forests, dinosaurs, flowering meadows, and finally us. Eight pages walk forward through geologic time, one period at a time; each shows what the plants and animals were like, what the atmosphere was doing, and what changed. Two closing pages step back to explain why it all happened: the slow buildup of oxygen and the ozone shield, the handful of evolutionary innovations that freed life from the water, and the volcanoes, drifting continents, and asteroid impacts that repeatedly reset the stage. Every content page has a voiceover you can play, images you can hover over to animate, and one graded question, ten in all.

The Exercise, Page by Page

Page	What you'll do
Introduction	A short video explains what to expect on the pages that follow.
1. The Pioneers	Ordovician–Silurian (500–420 Ma): the first life on land. One question.
2. The First Forests	Devonian (420–360 Ma): the first trees and the first four-legged animals. One question.
3. The Coal Forests	Carboniferous (360–299 Ma): giant plants, giant insects, and the oxygen peak. One question.
4. Supercontinent Pangaea	Permian (299–252 Ma): seed plants, synapsids, and the worst mass extinction. One question.
5. Recovery & Revolution	Triassic (252–201 Ma): the first dinosaurs and the first mammals. One question.
6. Age of Giant Dinosaurs	Jurassic (201–145 Ma): sauropods and the first birds. One question.
7. Flowering Plant Revolution	Cretaceous (145–66 Ma): flowers, pollinators, and the end of the dinosaurs. One question.
8. Age of Mammals & Humans	Cenozoic (66 Ma–present): grasslands, mammals, and Homo sapiens. One question.
9. The Atmospheric Foundation	How oxygen and the ozone layer made life on land possible. One question.
10. The Big Picture	The key innovations and the geologic forces behind them. One question.
Conclusion	A closing video, your score, and the Submit Quiz button.

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TRY IT IN THE EXERCISE

On each page, press Play to hear the voiceover, and hover over the images marked “hover to animate” to see them move. Keep the Numbers to Know table at hand as you go: most of the graded questions turn on a date, a percentage, or a single innovation.

Learning Objectives

After this exercise you should be able to:

- Explain why oxygen and the ozone layer had to come first before complex life could live on land.
- Put the major steps of plant evolution in order: crusts, vascular plants, trees, seed plants, and flowering plants.
- Describe how high Carboniferous oxygen made giant insects and other arthropods possible.
- Explain how the amniotic egg freed vertebrates from returning to water to reproduce.
- Identify the causes and the consequences of the End-Permian and End-Cretaceous mass extinctions.
- Describe the rise of dinosaurs, birds, and mammals through the Triassic, Jurassic, and Cretaceous.
- Explain the co-evolution of flowering plants and their insect pollinators.
- State when and where Homo sapiens evolved, and place it on the 500-million-year timeline.

QUICK REFERENCE

Key Terms

Term	Definition
Ma / Ga	Millions of years ago / billions of years ago. 430 Ma means 430 million years ago.
Cyanobacteria	Photosynthesizing microbes. They put the first oxygen into Earth's atmosphere and formed some of the earliest crusts on land.
Lichen	A partnership of a fungus with algae or cyanobacteria. Lichens grow on bare rock and slowly break it down into soil.
Bryophytes	Simple, low-growing plants without vascular tissue, such as mosses.
Mycorrhizae	Partnerships between fungi and plant roots; the fungus helps the plant take up water and nutrients.
Vascular tissue	A plant's internal plumbing: xylem carries water up, and phloem carries food around. It lets plants grow tall.
Tetrapod	A vertebrate with four limbs: amphibians, reptiles, birds, and mammals, and their ancestors.

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Term	Definition
Amniotic egg	A self-contained egg with its own water supply and a protective shell, so it can be laid on dry land.
Lycophytes	Club mosses. Today they are small, but in the Carboniferous some, such as Lepidodendron, were trees 50 m tall.
Gymnosperms	Seed plants whose seeds are not enclosed in fruit: conifers, cycads, and ginkgos.
Angiosperms	Flowering plants. Their seeds develop inside fruit.
Synapsids	The group of animals that includes mammals and their ancestors, such as the Permian gorgonopsids.
Archosaurs	The group that includes dinosaurs, pterosaurs, crocodilians, and birds.
Sauropods / Theropods	Sauropods: giant, long-necked, plant-eating dinosaurs. Theropods: two-legged, mostly meat-eating dinosaurs, the group that gave rise to birds.
Hominins	Humans and our extinct relatives after our lineage split from the other apes.
Pangaea	The supercontinent that held nearly all the land on Earth in the Permian and Triassic.
Mass extinction	A geologically brief event in which a large share of all species dies out.
Co-evolution	Two groups evolving together, each in response to the other, like flowers and their pollinators.
Great Oxidation Event	The time, about 2.4 Ga, when oxygen first began to build up in the atmosphere.
Ozone layer	A layer of ozone (a form of oxygen) high in the atmosphere that blocks the Sun's harmful ultraviolet (UV) light.

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QUICK REFERENCE

The Timeline

Period	When	Highlights
Ordovician–Silurian	500–420 Ma	Microbial and lichen crusts; mosses; Cooksonia, the first vascular plant; millipedes ashore
Devonian	420–360 Ma	First forests (Archaeopteris); deep roots make true soil; Tiktaalik; first tetrapods
Carboniferous	360–299 Ma	Coal swamps; oxygen peaks near 35%; giant insects; amniotic egg; first reptiles
Permian	299–252 Ma	Pangaea; gymnosperms and synapsids dominate; End-Permian extinction (252 Ma)
Triassic	252–201 Ma	Recovery; first dinosaurs (~230 Ma), pterosaurs, and mammals (~225 Ma)
Jurassic	201–145 Ma	Giant sauropods; conifer forests; Archaeopteryx (~150 Ma)
Cretaceous	145–66 Ma	Flowering plants and pollinators; T. rex; End-Cretaceous extinction (66 Ma)
Cenozoic	66 Ma–present	Mammals diversify; grasslands spread; Homo sapiens (~300,000 years ago)

QUICK REFERENCE

Numbers to Know

Value	What it refers to
500 million years	Span of the exercise, from the first crusts on land to today
2.4 Ga	Great Oxidation Event: oxygen begins to build up in the atmosphere
~430 Ma	Cooksonia, the first vascular plant, 2–6 cm tall
10–15%	Atmospheric oxygen when life first reached land
35%	Peak oxygen in the Carboniferous
21%	Oxygen in the atmosphere today
70 cm / 2.5 m	Wingspan of the dragonfly Meganeura / length of the millipede Arthropleura
50 m	Height of the Carboniferous scale tree Lepidodendron
~325 Ma	The amniotic egg appears
252 Ma, 96%	End-Permian extinction and the share of species it killed
66 Ma, 75%	End-Cretaceous extinction and the share of species it killed
10 km	Width of the Chicxulub asteroid
~300,000 years	Age of our species, Homo sapiens, which evolved in Africa

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EXERCISE CONTENT

Page-by-Page Summary

Part 1 — The Pioneers

Ordovician–Silurian, 500–420 Ma. The first life on land was microscopic. **Cyanobacteria** and **lichens** formed thin crusts on bare rock, and lichens began weathering that rock into the first soil. Simple mosses, or **bryophytes**, followed around 470 Ma, and by about 450 Ma fungi were forming partnerships with plant roots (**mycorrhizae**).

Around 430 Ma came **Cooksonia**, only 2–6 cm tall but the first plant with **vascular tissue**: internal plumbing to carry water. By about 428 Ma **millipedes** had ventured ashore, feeding on decaying plant material and needing moist surroundings to survive.

Oxygen was only about 10–15% of the atmosphere, compared with 21% today, but the ozone layer was strong enough to make the surface livable, if still harsh.

Part 2 — The First Forests

Devonian, 420–360 Ma. Plants grew taller. Club mosses (**lycophytes**) reached a meter, true ferns evolved complex leaves, and around 385 Ma the first forests appeared. Their best-known tree, **Archaeopteris**, grew up to 30 m tall.

Archaeopteris's great innovation was underground: roots more than a meter deep. They broke up rock and created **true soil**, and the faster weathering of rock changed the chemistry of the atmosphere and oceans.

In the water, **Tiktaalik** (375 Ma) was a “fishapod”: a fish with sturdy, limb-like fins, transitional between fish and four-limbed animals. By 365 Ma the first **tetrapods**, such as Acanthostega and Ichthyostega, had appeared. Spiders hunted in the undergrowth.

Part 3 — The Coal Forests

Carboniferous, 360–299 Ma. This was the age of giants. Scale trees (**Lepidodendron**) grew 50 m tall, tree ferns 20 m, and giant horsetails (Calamites) 18 m. These vast swamp forests were so productive that their buried, undecayed remains became today's **coal**.

Atmospheric **oxygen peaked at about 35%**, compared with 21% today. Insects and other arthropods take in oxygen by diffusion through their bodies, which normally limits how big they can grow; extra oxygen raised that limit. The dragonfly **Meganeura** had a 70 cm wingspan, and the millipede **Arthropleura** reached 2.5 m.

Around 325 Ma came the **amniotic egg**, an egg with its own water supply and a shell, which could be laid on dry land. It freed vertebrates from returning to water to reproduce, and the first reptiles and synapsids followed.

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Part 4 — Supercontinent Pangaea

Permian, 299–252 Ma. The continents had gathered into the supercontinent **Pangaea**, and its interior grew dry. **Gymnosperms** (conifers, cycads, ginkgos) and seed ferns such as Glossopteris took over, while spore plants like lycophytes and horsetails declined. Seeds did better than spores in dry environments.

Synapsids, the group that includes mammals' ancestors, ruled the land. **Gorgonopsids** were saber-toothed predators up to 3.5 m long, and large, barrel-chested plant-eaters roamed. The ancestors of dinosaurs and the first beetles appeared.

The period ended with the **End-Permian extinction** (252 Ma), the worst in Earth's history. Massive volcanic eruptions in the **Siberian Traps** released carbon dioxide, causing global warming, ocean acidification, and oxygen-starved (anoxic) oceans. About 96% of species died out.

Part 5 — Recovery & Revolution

Triassic, 252–201 Ma. Life recovered, but it looked very different. Gymnosperms still dominated the plants, and Pangaea's interior was extremely dry.

Dinosaurs first appeared around 230 Ma as small, two-legged predators such as Eoraptor and Herrerasaurus. Pterosaurs took to the skies around 228 Ma, and crocodilians diversified. Around 225 Ma the **first mammals** appeared: tiny, shrew-like, nocturnal insect-eaters.

Synapsids declined, except for the line leading to mammals, while the **archosaurs** — dinosaurs, pterosaurs, and crocodilians — rose to dominance, setting the stage for the Age of Dinosaurs.

Part 6 — Age of Giant Dinosaurs

Jurassic, 201–145 Ma. Gymnosperms reached their peak, with vast conifer forests over the continents and ferns in the understory. Carbon dioxide was 3–4 times today's level, keeping the climate warm and the vegetation lush.

Dinosaurs dominated. **Sauropods** such as Brachiosaurus and Diplodocus were the largest land animals that have ever lived. **Theropods** such as Allosaurus, about 9 m long, were the top predators, and Stegosaurus carried its famous back plates. Mammals stayed small and nocturnal.

Around 150 Ma **Archaeopteryx** appeared: the first known bird, with feathers and wings but also teeth, clawed fingers, and a bony tail. It marks the transition from theropod dinosaurs to birds.

Part 7 — The Flowering Plant Revolution

Cretaceous, 145–66 Ma. Flowering plants, or **angiosperms**, appeared around 140–130 Ma and diversified explosively; by the Late Cretaceous they dominated most environments. Magnolias are among the earliest, and have changed little since.

Their success came from **co-evolution with insects**. Flowers attracted pollinators such as bees, butterflies, and moths, and plants and pollinators diversified together. Fruits recruited animals to carry seeds away.

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Dinosaurs reached their peak diversity, including T. rex and Triceratops, while mammals and birds diversified. The **End-Cretaceous extinction** (66 Ma) was caused by the 10 km **Chicxulub asteroid** impact together with the Deccan volcanic eruptions. It ended the non-avian dinosaurs and the pterosaurs and killed about 75% of species, opening the door for mammals.

Part 8 — Age of Mammals & Humans

Cenozoic, 66 Ma to the present. Within about 10 million years of the dinosaurs' end, **mammals** had filled every major way of life. Primates appeared around 55 Ma and whales around 50 Ma, followed by horses, elephants, bats, and carnivores.

As the climate cooled, **grasslands** spread worldwide from around 23 Ma — savannas, prairies, and steppes — and grasses co-evolved with the grazing mammals that ate them.

Around 6–7 Ma the **hominin** line split from the other apes in Africa, and **Homo sapiens** evolved in Africa about 300,000 years ago. The ice ages of the last 2.6 million years brought mammoths, giant sloths, and saber-toothed cats, which thrived until humans arrived. Agriculture began about 11,700 years ago.

Part 9 — The Atmospheric Foundation

Nothing on the earlier pages could have happened without oxygen. **Cyanobacteria** began producing it by photosynthesis billions of years ago, and in the **Great Oxidation Event** (2.4 Ga) it began to build up in the atmosphere. From that oxygen, an **ozone layer** formed and began blocking the Sun's deadly ultraviolet (UV) light.

Without the ozone shield, UV light at ground level would be lethal to complex life. By about 470 Ma oxygen and ozone were sufficient for life to colonize the land, making them the prerequisite for all life on land.

Other forces shaped what came next. **Volcanic eruptions** released nutrients and carbon dioxide but also caused extinctions; **continental drift** assembled and broke up Pangaea, changing the climate; **asteroid impacts** cleared space for new groups; and ice ages and greenhouse periods forced life to adapt.

Part 10 — The Big Picture

Four innovations opened the land. **Vascular tissue** (430 Ma) let plants grow tall. **Seeds** protected plant embryos and freed plant reproduction from water. The **amniotic egg** (325 Ma) did the same for animals. **Flowers** (140 Ma), through co-evolution with pollinators, made angiosperms the most diverse plants on Earth.

Geologic forces drove the story too. Supercontinents formed dry interiors and, as they broke apart, created new and varied habitats. Mass extinctions — the End-Permian (252 Ma), which killed 96% of species, and the End-Cretaceous (66 Ma), which ended the dinosaurs — swept the stage clear, and each crisis created an opportunity for the survivors.

From microscopic crusts on bare rock to 50 m trees to human beings, the colonization of land took 500 million years, and it all started with oxygen.

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STUDY QUESTIONS

Questions for Review

Answer in your own words. Where a question asks you to explain rather than to state, a one-sentence answer is unlikely to be complete.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — The Pioneers

1. What were the earliest forms of life on land, and what did they do to the bare rock?

2. What is vascular tissue, and which plant was the first to have it?

3. Name the first animals to venture onto land. What did they eat, and what kind of environment did they need?

Part 2 — The First Forests

1. What was Archaeopteris, and how tall did it grow?

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2. Why were Archaeopteris's deep roots so important?

3. What was Tiktaalik, and why is it called a “fishapod”?

Part 3 — The Coal Forests

1. How high did atmospheric oxygen reach in the Carboniferous, and how does that compare with today?

2. Give two examples of giant Carboniferous animals, and explain how the high oxygen level made their size possible.

3. What is an amniotic egg, and why was it such an important innovation?

4. Where did today's coal come from?

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Part 4 — Supercontinent Pangaea

1. Which plants dominated the Permian, and why did seed plants do better than spore plants?

2. What were gorgonopsids?

3. What caused the End-Permian extinction, and how did it kill so many species?

Part 5 — Recovery & Revolution

1. When did the first dinosaurs appear, and what were they like?

2. When did the first mammals appear, and what were they like?

3. After the End-Permian extinction, which group declined and which group rose to dominance?

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Part 6 — Age of Giant Dinosaurs

1. What were sauropods? Name two.

2. What was Archaeopteryx, and when did it live?

3. Describe the Jurassic atmosphere and climate, and their effect on plant life.

Part 7 — The Flowering Plant Revolution

1. When did flowering plants appear, and what is their scientific name?

2. Explain how flowering plants and insects co-evolved, and what fruits added.

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3. What caused the End-Cretaceous extinction, and what did it make possible?

4. Why do scientists say the non-avian dinosaurs died out, rather than all the dinosaurs?

Part 8 — Age of Mammals & Humans

1. How quickly did mammals fill the roles left open by the dinosaurs? Name three groups that appeared.

2. When and why did grasslands spread, and how did they affect mammals?

3. When did the hominin line split from the other apes, and when and where did Homo sapiens evolve?

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Part 9 — The Atmospheric Foundation

1. What organisms first produced Earth's oxygen, and when did it begin to build up in the atmosphere?

2. Why was the ozone layer necessary before complex life could live on land?

3. Besides oxygen, name two geologic forces that drove the evolution of life on land, and say how.

Part 10 — The Big Picture

1. Name the four major innovations on the Big Picture page, and say what each one made possible.

2. How did the forming and breaking up of supercontinents affect evolution?

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3. "Each crisis created opportunity." Explain this using one of the two great mass extinctions.

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YOUR NOTES

Notes

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